

School of Computer Science and Engineering

Department of Computer Engineering and Technology

S. Y. B. Tech CSE

AY 2026-27

Semester 3

Computer Networks Lab (CSE40110)

Assignment No.: 5

Title: Write a program for TCP Socket Programming: (Language: C/C++/Python/Java).

Aim:

Write a C program for wired network using TCP socket to demonstrate

- Chat application (50% students)
- Mathematical operations. (Remaining 50% students)

Objectives:

- To understand the concept of socket programming.

Theory: (Handwritten & Scanned)

Students should write following theory

- What is Socket?
- TCP Socket Primitives at Server and Client

Program Code (for programming assignment) / CLI code (for Packet Tracer Assignments):

- Students should write/paste code here

Output Screenshots:

- Students should Draw/Paste final output screenshots

Students Observation:

- Students should write their observations about lab assignment

FAQ (Handwritten & Scanned)

- Students should write answer for following questions
 - What is a TCP Socket?
 - What are the main steps involved in TCP Socket Programming?
 - What is the role of IP address and port number in TCP Socket Programming?
 - Why is TCP called a connection-oriented protocol?
 - What is the difference between TCP and UDP socket primitives?